

Computing CMB temperature fluctuations for spherical spaces

Adriel Matei, Béla Schneider, Javier Vela, Juš Kocutar

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Manifolds & Homotopy groups

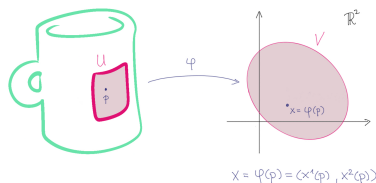


Figure: The prototypical example of a manifold a mug [13].

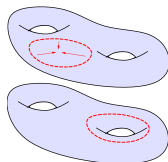


Figure: Two double tori with (non)-contractible paths [7].

Quotients of the 3-sphere

- $SO(4)$ is isomorphic to the isometry group of $\mathbb{S}^3$
- Subgroups $\Gamma \leq SO(4)$ define equivalence classes of orbits on $\mathbb{S}^3$ by the standard action of $SO(4)$ on $\mathbb{R}^3$, i.e. $x \sim y$ iff $x = My$ for some $M \in \Gamma$.
- The obtain space is (*sometimes*) a manifold. In particular, it is well defined and spherical for finite Γ .
- This can be easily generalised to the n -sphere.

Lens spaces

- Lens spaces are obtained by taking the quotient of some n -sphere by a cyclic group.
- They cannot be distinguished by their homotopy group alone.

Lens spaces — the explicit construction

Definition (Lens space)

Given $q \in \mathbb{N}$ and $s \in \mathbb{Z}^n$ each coprime with q

$$L(q, s) := \mathbb{S}^{2n-1} / \langle M_{q,s} \rangle,$$

where $\langle M \rangle$ is the group generated by

$$M_{q,s} := \begin{pmatrix} R(2\pi s_1/q) & & & \\ & R(2\pi s_2/q) & & \\ & & \ddots & \\ & & & R(2\pi s_n/q) \end{pmatrix}$$

In the $\mathbb{S}^3$ case, we denote $L(p, q) := L(p, (1, q))$ for coprime $p, q \in \mathbb{Z}$.

The Laplace-Beltrami operator

Definition (Laplace-Beltrami operator)

a generalization of the Laplace operator to more general spaces

$$\Delta f := \operatorname{div} \nabla f$$

- 1 The spectrum of this operator is a subset of $[0, \infty)$. We call it the *spectrum* of the manifold.
- 2 The spectrum determines many things about the space (like its volume).
- 3 When two distinct (up to isomorphism) manifolds share a spectrum, they form an *isospectral pair*.

The silver lining

- Isospectral pairs are cool and all, but they cannot occur in dimension 3.
- We can thus attempt to infer the shape of our universe based on its spectrum.

CMB Anisotropy of Homogeneous Spherical Spaces

- The Helmholtz equation on M has the spectrum as its solutions, and is given by

$$(\Delta + E_{\beta}^M)\psi_{\beta}^{M,i} = 0.$$

- In fact $E_{\beta}^m = \beta^2 - 1$ for $\beta \in \mathbb{N}$ we call β a wave number.
- The set of all possible wave numbers [for which there exists a non-zero solution] depends on Γ .
- Solutions to the Helmholtz equation are calculated for each Γ and CMB anisotropy $\frac{\delta T}{T}$ computed.

CMB Anisotropy of Homogeneous Spherical Spaces — Conclusion

- For the majority of groups Γ , the anisotropies $\frac{\delta T}{T}$ do not coincide with observations.
- The only groups for which it does are $\Gamma = O^*$ and $\Gamma = I^*$ — the **binary octahedral** and **binary icosahedral** groups of order 48 and 120 respectively.

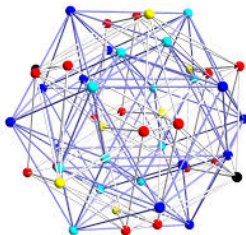


Figure: Graphical representation of the binary tetrahedral group [5]

Inhomogeneous spherical space

- Manifolds of the form $\mathbb{S}^3/\Gamma$ with Γ a group acting on $\mathbb{S}^3$.
- Multi-connected space: it has non-contractable loops
- Inhomogeneous space: it does not look identical from every point in space
- Fixing $|\Gamma| = 8$, we have three multi-connected manifolds, up to equivalence:
 - ① homogeneous: $N3$ and $L(8,1)$
 - ② inhomogeneous: $N2 \equiv L(8,3)$
- Results: inhomogeneous spaces have more variety in CMB anisotropies than homogeneous spaces, because the strength of anisotropy suppression is observer dependent.

The setup

From the perspective of an Earth-based observer, we can view CMB as a function defined on the celestial sphere $\mathbb{S}^2$. We expand $\Delta T(\hat{n})$ using *spherical harmonics*, yielding coefficients $a_{\ell m}$.

We assume the fluctuations:

- Statistically isotropic and homogeneous in the mean.
- Gaussian distribution.
- $a_{\ell m}$ are independent Gaussian variables.

Coefficients should satisfy: $\langle a_{\ell m} \rangle = 0$.

Remark: $\langle a_{\ell m} \rangle \neq 0$, suggests possible anisotropies or preferred directions in the universe

The setup

Real CMB observations are affected by instrumental noise and *sky masking*. As a result, estimating C_ℓ accurately requires simulations. Given C_ℓ^{th} , used to generate synthetic realizations of the CMB sky

$$a_{\ell m} \sim \mathcal{N}(0, C_\ell^{\text{th}}).$$

Unfortunately, this introduces bias, which must be corrected using a decorrelation matrix W , and a *sky mask* matrix M which accounts for issues caused by masking effects. This gives us the test statistic (tests the assumption of statistical isotropy):

$$S_i = \sum_j W_{ij} M_j,$$

which examines $\langle a_{\ell m} \rangle$ across multipole ranges.

Recall: $\langle a_{\ell m} \rangle \neq 0$, suggests possible anisotropies or preferred directions in the universe

The Results

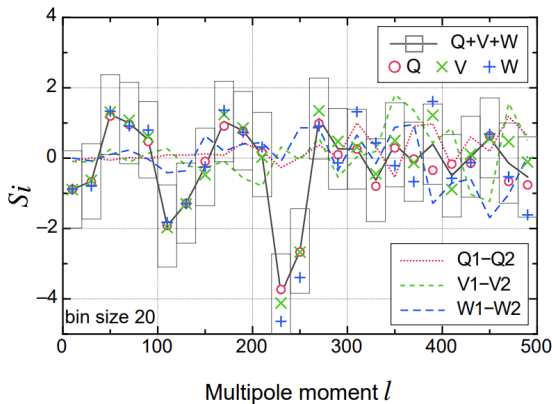


Figure: The decorrelated band mean spectrum obtained by the seven-year WMAP data [9].

Remark: In a positively curved space certain multipoles scales being preferred or suppressed in the CMB power spectrum.

Conclusion

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To summerize

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- 3

References

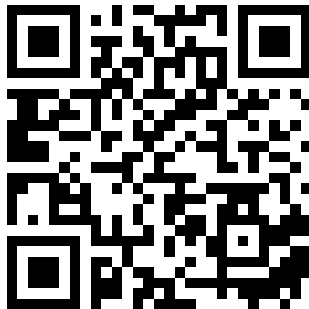
- R. Aurich, P. Kramer, and S. Lustig. *Cosmic microwave background radiation in an inhomogeneous spherical space*. Physica Scripta, 2011.
- R. Aurich, P. Kramer, and S. Lustig. *A survey of lens spaces and large-scale CMB anisotropy*. Monthly Notices of the Royal Astronomical Society, 2012.
- R. Aurich, S. Lustig, and F. Steiner. *CMB anisotropy of spherical spaces*. Classical and Quantum Gravity, 2005.
- P. Bielewicz, K. M. Górski, and A. J. Banday. *Low-order multipole maps of cosmic microwave background anisotropy derived from WMAP*. Oxford University Press, 2004.
- G. Egan. *Symmetries and the 24-cell*. 2021. Accessed at <https://www.gregegan.net/SCIENCE/24-cell/24-cell.html>.

References

- N. Jarosik et. al. *Seven-year Wilkinson Microwave Anisotropy Probe (WMAP) Observations: Sky Maps, Systematic Errors, and Basic Results*. The American Astronomical Society, 2011.
- M. Herlihy and N. Shavit. *The Topological Structure of Asynchronous Computability*. Journal of the ACM, 1996.
- A. Ikeda. *On the spectrum of a Riemannian manifold of positive constant curvature*. Osaka Journal of Mathematics, 1980.
- D. Kashino, K. Ichiki, and T. Takeuchi. *Test for anisotropy in the mean of the CMB temperature fluctuation in spherical harmonic space*. Physics Review, 2012.
- P. Labrana. *Emergent Universe Scenario and the Low CMB Multipoles*. Physical Review, 2013.

References

- E. A. Lauret and B. Linowitz. *The spectral geometry of hyperbolic and spherical manifolds: analogies and open problems*. New York Journal of Mathematics, 2025.
- R. Lehoucq, J. Weeks, J. P. Uzan, E. Gausmann, and J.P. Luminet. *Eigenmodes of three-dimensional spherical spaces and their application to cosmology*. Classical and Quantum Gravity, 2002.
- M. Serri. *Analysis on Manifolds*. AMS Open Math Notes, 2025.
- B. Tai-Dana, T. Bryson, and J. Terilla. *Topology, a categorical approach*. The MIT Press, 2020.



Thank You!